1. Decisions and regret

1. What is known on round t?

The feasible set $\mathcal{K} \subseteq \mathbb{R}^d$ is fixed, nonempty, compact and convex.

Each loss f_t is convex on $\mathcal{K}$ and differentiable on a neighborhood of it.

1. Choose $x_t \in \mathcal{K}$ using the past.
2. Observe f_t and incur $f_t(x_t)$.
3. Use the revealed loss to choose the next point.

The losses may change arbitrarily. The gradient at the point just played is $g_t = \nabla f_t(x_t)$.

2. One comparator for the whole sequence

Compare the learner with a single point $u \in \mathcal{K}$ used on every round. Its total loss is $\sum_{t=1}^{T} f_t(u)$. Regret compares the learner with the best such point on the same sequence.

$$R_T = \sum_{t=1}^{T} f_t(x_t) - \min_{u \in \mathcal{K}} \sum_{t=1}^{T} f_t(u).$$

Compactness and continuity ensure that a best fixed point exists.

Minimize **after** summing over rounds.

3. Linearize the loss gap

The tangent at x_t lies below the convex loss curve. For every $u \in \mathcal{K}$,

$$f_t(x_t) - f_t(u) \le g_t^T (x_t - u).$$

Both regret proofs use this inequality to turn loss differences into gradient inner products.

2. Feasible gradient steps

1. Euclidean projection

For nonempty closed convex $\mathcal{K}$, the closest feasible point to a proposal y is unique:

$$p = \Pi_{\mathcal{K}}(y) = \arg\min_{x \in \mathcal{K}} \|x - y\|_2.$$

For every $u \in \mathcal{K}$,

$$(y - p)^T (u - p) \le 0, \quad \|p - u\|_2 \le \|y - u\|_2.$$

The distance comparison starts at the **proposal** y. The previous played point can be closer to u than p is.

2. How to project

Box. Clip each coordinate to its allowed interval. On $[-1, 1]^2$, the proposal $(-1.3, 1.6)$ becomes $(-1, 1)$.

Ball. For a ball centered at zero with radius $B > 0$, keep y if $\|y\|_2 \le B$. Otherwise project to $B\frac{y}{\|y\|_2}$ on the boundary.

3. Projected online gradient descent (OGD)

Start with any $x_1 \in \mathcal{K}$. After round t, take a step opposite g_t and project:

$$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta_t g_t), \quad \eta_t > 0.$$

The current gradient first affects the decision on round $t + 1$. OGD needs only the gradient at the played point.

3. OGD regret by distance cancellation

1. One round and its sum

Fix $u \in \mathcal{K}$ and a constant $\eta > 0$. Let $a_t = \|x_t - u\|_2^2$. Expanding the squared distance after the gradient step and using projection gives

$$f_t(x_t) - f_t(u) \le \frac{a_t - a_{t+1}}{2\eta} + \frac{\eta}{2}\|g_t\|_2^2.$$

Sum before bounding the distances. The intermediate terms cancel, leaving

$$\sum_{t=1}^{T} (f_t(x_t) - f_t(u)) \le \frac{a_1 - a_{T+1}}{2\eta} + \frac{\eta}{2} \sum_{t=1}^{T} \|g_t\|_2^2.$$

2. Diameter, gradient bound and step size

Let D bound the diameter of $\mathcal{K}$ and assume $\|g_t\|_2 \le G$ at each played point. For $T \ge 1$,

$$R_T \le \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}.$$

For known T and known positive D, G, balance the terms:

$$\eta = \frac{D}{G\sqrt{T}}, \quad R_T \le DG\sqrt{T}.$$

If $D = 0$, there is only one decision. If $G = 0$, OGD stays at a common minimizer. Both give $R_T = 0$ with any positive step.

3. Squared loss on a parameter ball

For $f_t(x) = (a_t^T x - b_t)^2$ with $\|x\|_2 \le B$, $\|a_t\|_2 \le A$ and $|b_t| \le C$,

$$g_t = 2(a_t^T x_t - b_t) a_t.$$

The residual has magnitude at most $AB + C$. Use $D = 2B$ and $G = 2(AB + C)A$ in the OGD bound.

4. Leaders and regularization

1. Follow-the-leader (FTL)

From round 2, minimize the sum of the **original** past losses:

$$x_t \in \arg\min_{x\in\mathcal{K}} \sum_{s<t} f_s(x).$$

Specify x_1 and how ties are broken. Small changes in the sum can move the leader between opposite ends of an interval.

2. Linearized follow-the-regularized-leader

Assume $0 \in \mathcal{K}$ and use the quadratic regularizer $\mathcal{R}(x) = \alpha\frac{\|x\|_2^2}{2}$ with $\alpha > 0$. For fixed $\eta > 0$, quadratic FTRL plays $x_1 = 0$ and

$$x_t = \arg\min_{x\in\mathcal{K}} \left[\eta \sum_{s<t} g_s^T x + \mathcal{R}(x)\right].$$

Completing the square gives the update

$$x_t = \Pi_{\mathcal{K}}\left(-\frac{\eta}{\alpha}\sum_{s<t} g_s\right).$$

Each past gradient stays at its original value $g_s = \nabla f_s(x_s)$.

3. How much does a leader move?

On $[-1, 1]$, take linear losses $f_t(x) = g_t x$ with coefficients $\left(\frac{1}{2}, -1, 1, -1\right)$. The best fixed point $u = 1$ loses $-\frac{1}{2}$ in total.

Rule	Played points	Regret
FTL, start/tie at 1	$(1, -1, 1, -1)$	4
FTRL, $\alpha = 1$, $\eta = \frac{1}{2}$	$(0, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4})$	$\frac{5}{4}$

5. Quadratic FTRL regret by stability

1. Curvature bounds movement

For a differentiable α-strongly convex regularizer in Euclidean geometry,

$$\mathcal{R}(y) \geq \mathcal{R}(x) + \nabla\mathcal{R}(x)^T(y - x) + \frac{\alpha}{2}\|y - x\|_2^2.$$

Strong convexity can also hold at a corner, as in $|x| + \frac{x^2}{2}$. The differentiable quadratic used here meets the bound with equality. Adding a linear term preserves the quadratic gap. Comparing the minimizers before and after adding $\eta g_t^T x$ gives

$$\|x_t - x_{t+1}\|_2 \leq \frac{\eta}{\alpha}\|g_t\|_2.$$

2. Compare with the fixed point

The next point x_{t+1} already minimizes the objective that includes g_t. Comparing these prefix minimizers with $u \in \mathcal{K}$ yields

$$\sum_{t=1}^{T} g_t^T(x_{t+1} - u) \leq \frac{\mathcal{R}(u)}{\eta}.$$

The cost of using the played point x_t on that round is

$$g_t^T(x_t - x_{t+1}) \leq \frac{\eta}{\alpha}\|g_t\|_2^2.$$

3. The quadratic FTRL bound

With $\|g_t\|_2 \leq G$ on all played points,

$$\sum_{t=1}^{T}(f_t(x_t) - f_t(u)) \leq \frac{\mathcal{R}(u)}{\eta} + \frac{\eta G^2 T}{\alpha}.$$

Choose a known B with $\mathcal{R}(u) \leq B$ for **all** feasible u. For known T and positive B, G,

$$\eta = \sqrt{\frac{\alpha B}{G^2 T}}, \quad R_T \leq 2G\sqrt{\frac{BT}{\alpha}}.$$

For instance, $0 \in \mathcal{K}$ and diameter at most D allow $B = \alpha\frac{D^2}{2}$. If $B = 0$ or $G = 0$, regret is zero for any positive step.

6. Interpreting the guarantees

1. When do the two updates agree?

Without constraints, quadratic FTRL from zero takes gradient steps of size $\frac{\eta}{\alpha}$. Under constraints, OGD projects each step from the previous point. FTRL projects the scaled sum of past gradients.

On $[-1, 1]$, start at 0 with $\eta = \alpha = 1$ and gradients $(2, -1)$. Both reach $x_2 = -1$. The next OGD step gives $x_3 = 0$, while FTRL gives $x_3 = -1$.

2. Worst-case regret on a sequence

On an interval of diameter $D > 0$ with gradient bound $G > 0$, every online algorithm can face expected regret of at least $cDG\sqrt{T}$ for a universal $c > 0$. Linear losses already suffice.

This matches tuned OGD in worst-case order. A quadratic regularizer alone cannot give logarithmic regret for general convex losses.

3. Expected loss on a fresh draw

Now suppose losses are independent draws from one distribution, $F(x) = \mathbb{E}[f_t(x)]$ is finite, and x_t uses only the past. Under the OGD assumptions and tuned step on page 1, return the average

$$\bar{x}_T = \frac{1}{T}\sum_{t=1}^{T} x_t \in \mathcal{K}.$$

The expected excess population loss satisfies

$$\mathbb{E}[F(\bar{x}_T)] - \min_{u\in\mathcal{K}} F(u) = O\left(\frac{DG}{\sqrt{T}}\right).$$

Fresh draws relate expected round losses to F. Convexity lets us use the average decision. The last point x_T has no such guarantee here.